

SIEMENS MAGNETOM TrioTim syngo MR B19

\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\localizer

TA: 0:30 PAT: Off Voxel size: 1.7×1.3×7.0 mm Rel. SNR: 1.00 SIEMENS: gre

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	On
Load images to graphic segments	Off
Auto open inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	single

Routine

Slice group 1	
Slices	5
Dist. factor	200 %
Position	L0.0 A20.0 H0.0
Orientation	Coronal
Phase enc. dir.	R >> L
Rotation	0.00 deg
Slice group 2	
Slices	3
Dist. factor	200 %
Position	R70.0 P0.0 H0.0
Orientation	Sagittal
Phase enc. dir.	A >> P
Rotation	0.00 deg
Slice group 3	
Slices	3
Dist. factor	200 %
Position	L70.0 P0.0 H0.0
Orientation	Sagittal
Phase enc. dir.	A >> P
Rotation	0.00 deg
Slice group 4	
Slices	5
Dist. factor	200 %
Position	Isocenter
Orientation	Transversal
Phase enc. dir.	A >> P
Rotation	-0.20 deg
Phase oversampling	0 %
FoV read	400 mm
FoV phase	100.0 %
Slice thickness	7.0 mm
TR	7.4 ms
TE	3.01 ms
Averages	1
Concatenations	16
Filter	Elliptical filter
Coil elements	LBR;RBR

Contrast

TD	0 ms
MTC	Off
Magn. preparation	None
Flip angle	20 deg
Fat suppr.	None
Water suppr.	None
SWI	Off

Averaging mode

Reconstruction	Short term
Measurements	Magnitude
Multiple series	1
	Each measurement

Resolution

Base resolution	320
Phase resolution	75 %
Phase partial Fourier	Off
Interpolation	Off
PAT mode	None
Matrix Coil Mode	Auto (CP)
Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	Off
Normalize	Off
B1 filter	Off
Raw filter	Off
Elliptical filter	On
Mode	Inplane

Geometry

Multi-slice mode	Sequential
Series	Interleaved
Saturation mode	Standard
Special sat.	None
Tim CT mode	Off

System

Body	Off
LBR	On
RBR	On
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Adaptive Combine
Auto Coil Select	Default
Shim mode	Tune up
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
R >> L	350 mm
A >> P	263 mm
F >> H	350 mm

Physio

1st Signal/Mode	None
Segments	1
Dark blood	Off
Resp. control	Off

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Inline

Subtract	Off
Liver registration	Off
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On
Wash - In	Off
Wash - Out	Off
TTP	Off
PEI	Off
MIP - time	Off

Sequence

Introduction	On
Dimension	2D
Phase stabilisation	Off
Asymmetric echo	Off
Contrasts	1
Bandwidth	320 Hz/Px
Flow comp.	No
Allowed delay	0 s
RF pulse type	Fast
Gradient mode	Normal
Excitation	Slice-sel.
RF spoiling	On

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\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\STIR COR if known CANCER

TA: 5:22 PAT: Off Voxel size: 1.4×1.4×5.0 mm Rel. SNR: 1.00 SIEMENS: tse

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	On
Load images to graphic segments	Off
Auto open inline display	Off
Start measurement without further preparation	On
Wait for user to start	Off
Start measurements	single

Routine

Slice group 1	
Slices	32
Dist. factor	20 %
Position	L5.7 P110.7 H1.0
Orientation	Coronal
Phase enc. dir.	F >> H
Rotation	90.00 deg
Phase oversampling	50 %
FoV read	450 mm
FoV phase	100.0 %
Slice thickness	5.0 mm
TR	6670 ms
TE	95.0 ms
Averages	1
Concatenations	2
Filter	Elliptical filter
Coil elements	BC

Contrast

TD	0.0 ms
MTC	Off
Magn. preparation	Slice-sel. IR
TI	230 ms
Freeze suppressed tissue	Off
Flip angle	120 deg
Fat suppr.	None
Water suppr.	None
Restore magn.	Off
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution

Base resolution	320
Phase resolution	98 %
Phase partial Fourier	Off
Trajectory	Cartesian
Interpolation	Off
PAT mode	None
Matrix Coil Mode	Auto (CP)
Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	Off
Normalize	Off
B1 filter	Off

Raw filter	Off
Elliptical filter	On
Mode	Inplane

Geometry

Multi-slice mode	Interleaved
Series	Interleaved
Special sat.	None
Tim CT mode	Off

System

Body	On
LBR	Off
RBR	Off
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Adaptive Combine
Auto Coil Select	Default
Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	L5.7 P110.7 H1.0
Orientation	Coronal
Rotation	90.00 deg
R >> L	450 mm
F >> H	450 mm
A >> P	191 mm

Physio

1st Signal/Mode	None
Dark blood	Off
Resp. control	Off

Inline

Subtract	Off
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Sequence

Introduction	On
Dimension	2D
Compensate T2 decay	Off
Reduce Motion Sens.	Off
Contrasts	1
Bandwidth	300 Hz/Px

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Flow comp.	No
Allowed delay	0 s
Echo spacing	8.68 ms
Define	Turbo factor
Turbo factor	21
Echo trains per slice	23
RF pulse type	Low SAR
Gradient mode	Fast

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\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\t1_fl3d_tra_nonfatsat

TA: 1:28 PAT: 2 Voxel size: 1.1×0.8×1.0 mm Rel. SNR: 1.00 SIEMENS: gre

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Start measurement without further preparation	On
Wait for user to start	Off
Start measurements	single

Routine

Slab group 1	
Slabs	1
Dist. factor	20 %
Position	R2.9 P0.0 F20.2
Orientation	Transversal
Phase enc. dir.	R >> L
Rotation	90.00 deg
Phase oversampling	27 %
Slice oversampling	11.1 %
Slices per slab	144
FoV read	340 mm
FoV phase	100.0 %
Slice thickness	1.00 mm
TR	6.5 ms
TE	2.60 ms
Averages	1
Concatenations	1
Filter	Elliptical filter
Coil elements	LBR;RBR

Contrast

MTC	Off
Magn. preparation	None
Flip angle	25 deg
Fat suppr.	None
Water suppr.	None
SWI	Off
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution

Base resolution	448
Phase resolution	72 %
Slice resolution	66 %
Phase partial Fourier	6/8
Slice partial Fourier	6/8
Interpolation	Off
PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	24
Matrix Coil Mode	Auto (Triple)
Reference scan mode	Integrated
Image Filter	Off
Distortion Corr.	Off

Prescan Normalize	Off
Normalize	Off
B1 filter	Off
Raw filter	Off
Elliptical filter	On
Mode	Inplane

Geometry

Multi-slice mode	Interleaved
Series	Interleaved
Saturation mode	Standard
Special sat.	None
Tim CT mode	Off

System

Body	Off
LBR	On
RBR	On
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Adaptive Combine
Auto Coil Select	Default
Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	R2.9 P0.0 F20.2
Orientation	Transversal
Rotation	90.00 deg
A >> P	340 mm
R >> L	340 mm
F >> H	144 mm

Physio

1st Signal/Mode	None
Segments	1
Dark blood	Off
Resp. control	Off

Inline

Subtract	Off
Liver registration	Off
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On
Wash - In	Off

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Wash - Out	Off
TTP	Off
PEI	Off
MIP - time	Off

Sequence

Introduction	On
Dimension	3D
Elliptical scanning	Off
Phase stabilisation	Off
Asymmetric echo	Allowed
Contrasts	1
Bandwidth	470 Hz/Px
Flow comp.	No
Allowed delay	0 s
RF pulse type	Normal
Gradient mode	Fast
Excitation	Slab-sel.
RF spoiling	On

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\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\t2_tse_tra_spair

TA: 3:09 PAT: 2 Voxel size: 0.9×0.7×3.5 mm Rel. SNR: 1.00 SIEMENS: tse

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Start measurement without further preparation	On
Wait for user to start	Off
Start measurements	single

Unfiltered images	Off
Distortion Corr.	Off
Prescan Normalize	Off
Normalize	Off
B1 filter	Off
Raw filter	Off
Elliptical filter	On
Mode	Inplane

Geometry

Multi-slice mode	Interleaved
Series	Interleaved
Special sat.	None
Tim CT mode	Off

Routine

Slice group 1	
Slices	40
Dist. factor	20 %
Position	L5.7 P40.7 H0.0
Orientation	Transversal
Phase enc. dir.	R >> L
Rotation	90.00 deg
Phase oversampling	20 %
FoV read	340 mm
FoV phase	100.0 %
Slice thickness	3.5 mm
TR	7800 ms
TE	84 ms
Averages	1
Concatenations	1
Filter	Elliptical filter, Image Filter
Coil elements	LBR;RBR

System

Body	Off
LBR	On
RBR	On
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Adaptive Combine
Auto Coil Select	Default
Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	On
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	L5.7 P40.7 H0.0
Orientation	Transversal
Rotation	90.00 deg
A >> P	340 mm
R >> L	340 mm
F >> H	168 mm

Contrast

MTC	Off
Magn. preparation	None
Flip angle	80 deg
Fat suppr.	SPAIR
Fat sat. mode	Strong
Water suppr.	None
Restore magn.	Off
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Physio

1st Signal/Mode	None
Dark blood	Off
Resp. control	Off

Resolution

Base resolution	512
Phase resolution	75 %
Phase partial Fourier	Off
Trajectory	Cartesian
Interpolation	Off
PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	43
Matrix Coil Mode	Auto (Triple)
Reference scan mode	Integrated
Image Filter	On
Intensity	Medium
Edge Enhancement	3
Smoothing	3

Inline

Subtract	Off
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Sequence

Introduction	On
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Dimension	2D
Compensate T2 decay	Off
Reduce Motion Sens.	Off
Contrasts	1
Bandwidth	305 Hz/Px
Flow comp.	No
Allowed delay	0 s
Echo spacing	11.9 ms
Define	Turbo factor
Turbo factor	11
Echo trains per slice	23
RF pulse type	Normal
Gradient mode	Fast

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\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\SIEMENS t1_3d_tra_90s-dynFS - TEST

TA: 1:33 PAT: 2 Voxel size: 1.0x0.8x0.9 mm Rel. SNR: 1.00 SIEMENS: fl3d_vibe

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Start measurement without further preparation	On
Wait for user to start	On
Start measurements	single

Routine

Slab group 1	
Slabs	1
Dist. factor	20 %
Position	L0.0 P22.9 H1.7
Orientation	Transversal
Phase enc. dir.	R >> L
Rotation	90.00 deg
Phase oversampling	7 %
Slice oversampling	30.0 %
Slices per slab	160
FoV read	340 mm
FoV phase	100.0 %
Slice thickness	0.90 mm
TR	4.10 ms
TE	1.45 ms
Averages	1
Concatenations	1
Filter	Elliptical filter
Coil elements	LBR;RBR

Contrast

Flip angle	10.0 deg
Fat suppr.	SPAIR
Lines Per Shot	40
Water suppr.	None
Dixon	No Dixon
Save original images	On
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution

Base resolution	416
Phase resolution	82 %
Slice resolution	61 %
Phase partial Fourier	6/8
Slice partial Fourier	6/8
Interpolation	Off
PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	24
Matrix Coil Mode	Auto (Triple)
Reference scan mode	Integrated
Image Filter	Off
Distortion Corr.	Off

Prescan Normalize	Off
Normalize	Off
B1 filter	Off
Raw filter	Off
Elliptical filter	On
Mode	Inplane
POCS	Off

Geometry

Multi-slice mode	Sequential
Series	Ascending
Special sat.	None

System

Body	Off
LBR	On
RBR	On
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Adaptive Combine
Auto Coil Select	Default
Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	On
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	L0.0 P22.9 H1.7
Orientation	Transversal
Rotation	90.00 deg
A >> P	340 mm
R >> L	340 mm
F >> H	144 mm

Physio

1st Signal/Mode	None
Resp. control	Off

Inline

3D centric reordering	Off
Subtract	On
Liver registration	Off
Autoscaling	On
Subtrahend	1
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	On
MIP-Time	Off
Save original images	On
Wash - In	Off
Wash - Out	Off

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TTP	Off
PEI	Off
MIP - time	Off

Sequence

Introduction	Off
Dimension	3D
Elliptical scanning	Off
Reduce Motion Sens.	On
Asymmetric echo	Weak
Contrasts	1
Bandwidth	450 Hz/Px
Optimization	Min. TE
Allowed delay	0 s
RF pulse type	Normal
Gradient mode	Fast
Excitation	Slab-sel.
RF spoiling	On

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\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\SIEMENS t1_3d_tra_90s-dynFS - TEST

TA: 1:33 PAT: 2 Voxel size: 1.0x0.8x0.9 mm Rel. SNR: 1.00 SIEMENS: fl3d_vibe

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Start measurement without further preparation	On
Wait for user to start	On
Start measurements	single

Routine

Slab group 1	
Slabs	1
Dist. factor	20 %
Position	L0.0 P22.9 H1.7
Orientation	Transversal
Phase enc. dir.	R >> L
Rotation	90.00 deg
Phase oversampling	7 %
Slice oversampling	30.0 %
Slices per slab	160
FoV read	340 mm
FoV phase	100.0 %
Slice thickness	0.90 mm
TR	4.10 ms
TE	1.45 ms
Averages	1
Concatenations	1
Filter	Elliptical filter
Coil elements	LBR;RBR

Contrast

Flip angle	10.0 deg
Fat suppr.	SPAIR
Lines Per Shot	40
Water suppr.	None
Dixon	No Dixon
Save original images	On
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution

Base resolution	416
Phase resolution	82 %
Slice resolution	61 %
Phase partial Fourier	6/8
Slice partial Fourier	6/8
Interpolation	Off
PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	24
Matrix Coil Mode	Auto (Triple)
Reference scan mode	Integrated
Image Filter	Off
Distortion Corr.	Off

Prescan Normalize	Off
Normalize	Off
B1 filter	Off
Raw filter	Off
Elliptical filter	On
Mode	Inplane
POCS	Off

Geometry

Multi-slice mode	Sequential
Series	Ascending
Special sat.	None

System

Body	Off
LBR	On
RBR	On
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Adaptive Combine
Auto Coil Select	Default
Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	On
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	L0.0 P22.9 H1.7
Orientation	Transversal
Rotation	90.00 deg
A >> P	340 mm
R >> L	340 mm
F >> H	144 mm

Physio

1st Signal/Mode	None
Resp. control	Off

Inline

3D centric reordering	Off
Subtract	On
Liver registration	Off
Autoscaling	On
Subtrahend	1
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	On
MIP-Time	Off
Save original images	On
Wash - In	Off
Wash - Out	Off

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TTP	Off
PEI	Off
MIP - time	Off

Sequence

Introduction	Off
Dimension	3D
Elliptical scanning	Off
Reduce Motion Sens.	On
Asymmetric echo	Weak
Contrasts	1
Bandwidth	450 Hz/Px
Optimization	Min. TE
Allowed delay	0 s
RF pulse type	Normal
Gradient mode	Fast
Excitation	Slab-sel.
RF spoiling	On

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\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\SIEMENS
TA: 6:31 PAT: 2 Voxel size: 1.0x0.8x0.9 mm Rel. SNR: 1.00 SIEMENS: fl3d_vibe

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Start measurement without further preparation	On
Wait for user to start	On
Start measurements	single

Image Filter	Off
Distortion Corr.	Off
Prescan Normalize	Off
Normalize	Off
B1 filter	Off
Raw filter	Off
Elliptical filter	On
Mode	Inplane
POCS	Off

Geometry

Multi-slice mode	Sequential
Series	Ascending
Special sat.	None

Routine

Slab group 1	
Slabs	1
Dist. factor	20 %
Position	L0.0 P22.9 H1.7
Orientation	Transversal
Phase enc. dir.	R >> L
Rotation	90.00 deg
Phase oversampling	7 %
Slice oversampling	30.0 %
Slices per slab	160
FoV read	340 mm
FoV phase	100.0 %
Slice thickness	0.90 mm
TR	4.10 ms
TE	1.45 ms
Averages	1
Concatenations	1
Filter	Elliptical filter
Coil elements	LBR;RBR

System

Body	Off
LBR	On
RBR	On
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Adaptive Combine
Auto Coil Select	Default
Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	On
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	L0.0 P22.9 H1.7
Orientation	Transversal
Rotation	90.00 deg
A >> P	340 mm
R >> L	340 mm
F >> H	144 mm

Contrast

Flip angle	10.0 deg
Fat suppr.	SPAIR
Lines Per Shot	40
Water suppr.	None
Dixon	No Dixon
Save original images	On
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	4
Pause after meas. 1	20.0 s
Pause after meas. 2	0.0 s
Pause after meas. 3	0.0 s
Multiple series	Each measurement

Physio

1st Signal/Mode	None
Resp. control	Off

Inline

3D centric reordering	Off
Subtract	On
Liver registration	Off
Autoscaling	On
Subtrahend	1
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	On
MIP-Time	Off
Save original images	On

Resolution

Base resolution	416
Phase resolution	82 %
Slice resolution	61 %
Phase partial Fourier	6/8
Slice partial Fourier	6/8
Interpolation	Off
PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	24
Matrix Coil Mode	Auto (Triple)
Reference scan mode	Integrated

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Wash - In	Off
Wash - Out	Off
TTP	Off
PEI	Off
MIP - time	Off

Sequence

Introduction	Off
Dimension	3D
Elliptical scanning	Off
Reduce Motion Sens.	On
Asymmetric echo	Weak
Contrasts	1
Bandwidth	450 Hz/Px
Optimization	Min. TE
Allowed delay	0 s
RF pulse type	Normal
Gradient mode	Fast
Excitation	Slab-sel.
RF spoiling	On

SIEMENS MAGNETOM TrioTim syngo MR B19

\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\T2fs SAG RIGHT

TA: 5:12 PAT: Off Voxel size: 0.6×0.6×3.0 mm Rel. SNR: 1.00 SIEMENS: tse

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	On
Load images to graphic segments	On
Auto open inline display	Off
Start measurement without further preparation	On
Wait for user to start	Off
Start measurements	single

Routine

Slice group 1	
Slices	43
Dist. factor	0 %
Position	R67.1 A1.0 H1.9
Orientation	Sagittal
Phase enc. dir.	H >> F
Rotation	90.00 deg
Phase oversampling	100 %
FoV read	180 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	4700 ms
TE	127.0 ms
Averages	1
Concatenations	2
Filter	Prescan Normalize, Elliptical filter
Coil elements	RBR

Contrast

TD	0.0 ms
MTC	Off
Magn. preparation	None
Flip angle	150 deg
Fat suppr.	Fat sat.
Fat sat. mode	Strong
Water suppr.	None
Restore magn.	Off
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution

Base resolution	320
Phase resolution	100 %
Phase partial Fourier	Off
Trajectory	Cartesian
Interpolation	Off
PAT mode	None
Matrix Coil Mode	Auto (CP)
Image Filter	Off
Distortion Corr.	Off
Unfiltered images	Off
Prescan Normalize	On
Normalize	Off

B1 filter	Off
Raw filter	Off
Elliptical filter	On
Mode	Inplane

Geometry

Multi-slice mode	Interleaved
Series	Interleaved
Special sat.	None
Tim CT mode	Off

System

Body	Off
LBR	Off
RBR	On
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Sum of Squares
Auto Coil Select	Default
Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	On
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	R67.1 A1.0 H1.9
Orientation	Sagittal
Rotation	90.00 deg
A >> P	180 mm
F >> H	180 mm
R >> L	129 mm

Physio

1st Signal/Mode	None
Dark blood	Off
Resp. control	Off

Inline

Subtract	Off
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Sequence

Introduction	On
Dimension	2D
Compensate T2 decay	Off
Reduce Motion Sens.	Off
Contrasts	1

SIEMENS MAGNETOM TrioTim syngo MR B19

Bandwidth	300 Hz/Px
Flow comp.	No
Allowed delay	30 s
Echo spacing	9.1 ms
Define	Turbo factor
Turbo factor	21
Echo trains per slice	31
RF pulse type	Low SAR
Gradient mode	Fast

SIEMENS MAGNETOM TrioTim syngo MR B19

\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\T2fs SAG LEFT

TA: 5:12 PAT: Off Voxel size: 0.6×0.6×3.0 mm Rel. SNR: 1.00 SIEMENS: tse

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	On
Load images to graphic segments	On
Auto open inline display	Off
Start measurement without further preparation	On
Wait for user to start	Off
Start measurements	single

Routine

Slice group 1	
Slices	43
Dist. factor	0 %
Position	L67.1 A1.0 H1.9
Orientation	Sagittal
Phase enc. dir.	H >> F
Rotation	90.00 deg
Phase oversampling	100 %
FoV read	180 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	4700 ms
TE	127.0 ms
Averages	1
Concatenations	2
Filter	Prescan Normalize, Elliptical filter
Coil elements	LBR

Contrast

TD	0.0 ms
MTC	Off
Magn. preparation	None
Flip angle	150 deg
Fat suppr.	Fat sat.
Fat sat. mode	Strong
Water suppr.	None
Restore magn.	Off
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution

Base resolution	320
Phase resolution	100 %
Phase partial Fourier	Off
Trajectory	Cartesian
Interpolation	Off
PAT mode	None
Matrix Coil Mode	Auto (CP)
Image Filter	Off
Distortion Corr.	Off
Unfiltered images	Off
Prescan Normalize	On
Normalize	Off

B1 filter	Off
Raw filter	Off
Elliptical filter	On
Mode	Inplane

Geometry

Multi-slice mode	Interleaved
Series	Interleaved
Special sat.	None
Tim CT mode	Off

System

Body	Off
LBR	On
RBR	Off
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Sum of Squares
Auto Coil Select	Default
Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	On
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	L67.1 A1.0 H1.9
Orientation	Sagittal
Rotation	90.00 deg
A >> P	180 mm
F >> H	180 mm
R >> L	129 mm

Physio

1st Signal/Mode	None
Dark blood	Off
Resp. control	Off

Inline

Subtract	Off
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Sequence

Introduction	On
Dimension	2D
Compensate T2 decay	Off
Reduce Motion Sens.	Off
Contrasts	1

SIEMENS MAGNETOM TrioTim syngo MR B19

Bandwidth	300 Hz/Px
Flow comp.	No
Allowed delay	30 s
Echo spacing	9.1 ms
Define	Turbo factor
Turbo factor	21
Echo trains per slice	31
RF pulse type	Low SAR
Gradient mode	Fast

SIEMENS MAGNETOM TrioTim syngo MR B19

\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\T2 SAG RIGHT non fs

TA: 4:42

PAT: Off

Voxel size: 0.6×0.6×2.5 mm

Rel. SNR: 1.00

SIEMENS: tse

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	On
Load images to graphic segments	On
Auto open inline display	Off
Start measurement without further preparation	On
Wait for user to start	Off
Start measurements	single

Routine

Slice group 1	
Slices	43
Dist. factor	0 %
Position	R67.1 A1.0 H1.9
Orientation	Sagittal
Phase enc. dir.	H >> F
Rotation	90.00 deg
Phase oversampling	100 %
FoV read	200 mm
FoV phase	100.0 %
Slice thickness	2.5 mm
TR	4390 ms
TE	129.0 ms
Averages	1
Concatenations	2
Filter	Distortion Corr.(2D), Prescan
Coil elements	Normalize, Elliptical filter
	RBR

Contrast

TD	0.0 ms
MTC	Off
Magn. preparation	None
Flip angle	150 deg
Fat suppr.	None
Water suppr.	None
Restore magn.	Off
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution

Base resolution	320
Phase resolution	100 %
Phase partial Fourier	Off
Trajectory	Cartesian
Interpolation	Off
PAT mode	None
Matrix Coil Mode	Auto (CP)
Image Filter	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off
Unfiltered images	Off
Prescan Normalize	On

Normalize	Off
B1 filter	Off
Raw filter	Off
Elliptical filter	On
Mode	Inplane

Geometry

Multi-slice mode	Interleaved
Series	Interleaved
Special sat.	None
Tim CT mode	Off

System

Body	Off
LBR	Off
RBR	On
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Sum of Squares
Auto Coil Select	Default
Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	R67.1 A1.0 H1.9
Orientation	Sagittal
Rotation	90.00 deg
A >> P	200 mm
F >> H	200 mm
R >> L	108 mm

Physio

1st Signal/Mode	None
Dark blood	Off
Resp. control	Off

Inline

Subtract	Off
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Sequence

Introduction	On
Dimension	2D
Compensate T2 decay	Off
Reduce Motion Sens.	Off

SIEMENS MAGNETOM TrioTim syngo MR B19

Contrasts	1
Bandwidth	300 Hz/Px
Flow comp.	No
Allowed delay	30 s
Echo spacing	9.18 ms
Define	Turbo factor
Turbo factor	21
Echo trains per slice	31
RF pulse type	Low SAR
Gradient mode	Fast

SIEMENS MAGNETOM TrioTim syngo MR B19

\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\T2 SAG LEFT non fs

TA: 4:42 PAT: Off Voxel size: 0.6×0.6×2.5 mm Rel. SNR: 1.00 SIEMENS: tse

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	On
Load images to graphic segments	On
Auto open inline display	Off
Start measurement without further preparation	On
Wait for user to start	Off
Start measurements	single

Routine

Slice group 1	
Slices	43
Dist. factor	0 %
Position	L67.1 A1.0 H1.9
Orientation	Sagittal
Phase enc. dir.	H >> F
Rotation	90.00 deg
Phase oversampling	100 %
FoV read	200 mm
FoV phase	100.0 %
Slice thickness	2.5 mm
TR	4390 ms
TE	129.0 ms
Averages	1
Concatenations	2
Filter	Distortion Corr.(2D), Prescan
Coil elements	Normalize, Elliptical filter
	LBR

Contrast

TD	0.0 ms
MTC	Off
Magn. preparation	None
Flip angle	150 deg
Fat suppr.	None
Water suppr.	None
Restore magn.	Off
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution

Base resolution	320
Phase resolution	100 %
Phase partial Fourier	Off
Trajectory	Cartesian
Interpolation	Off
PAT mode	None
Matrix Coil Mode	Auto (CP)
Image Filter	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off
Unfiltered images	Off
Prescan Normalize	On

Normalize	Off
B1 filter	Off
Raw filter	Off
Elliptical filter	On
Mode	Inplane

Geometry

Multi-slice mode	Interleaved
Series	Interleaved
Special sat.	None
Tim CT mode	Off

System

Body	Off
LBR	On
RBR	Off
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Sum of Squares
Auto Coil Select	Default
Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	L67.1 A1.0 H1.9
Orientation	Sagittal
Rotation	90.00 deg
A >> P	200 mm
F >> H	200 mm
R >> L	108 mm

Physio

1st Signal/Mode	None
Dark blood	Off
Resp. control	Off

Inline

Subtract	Off
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Sequence

Introduction	On
Dimension	2D
Compensate T2 decay	Off
Reduce Motion Sens.	Off

SIEMENS MAGNETOM TrioTim syngo MR B19

Contrasts	1
Bandwidth	300 Hz/Px
Flow comp.	No
Allowed delay	30 s
Echo spacing	9.18 ms
Define	Turbo factor
Turbo factor	21
Echo trains per slice	31
RF pulse type	Low SAR
Gradient mode	Fast

SIEMENS MAGNETOM TrioTim syngo MR B19

\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\+C 3D T1 sag RIGHT

TA: 3:42 PAT: 2 Voxel size: 0.7×0.6×1.0 mm Rel. SNR: 1.00 SIEMENS: fl3d_vibe

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Start measurement without further preparation	On
Wait for user to start	On
Start measurements	single

Distortion Corr.	Off
Unfiltered images	Off
Prescan Normalize	On
Normalize	Off
B1 filter	Off
Raw filter	Off
Elliptical filter	On
Mode	Inplane
POCS	Off

Geometry

Multi-slice mode	Sequential
Series	Ascending
Special sat.	None

System

Body	Off
LBR	Off
RBR	On
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Sum of Squares
Auto Coil Select	Default
Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	On
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	R85.9 A22.9 H17.4
Orientation	Sagittal
Rotation	90.00 deg
A >> P	180 mm
F >> H	180 mm
R >> L	144 mm

Routine

Slab group 1	
Slabs	1
Dist. factor	20 %
Position	R85.9 A22.9 H17.4
Orientation	Sagittal
Phase enc. dir.	H >> F
Rotation	90.00 deg
Phase oversampling	100 %
Slice oversampling	22.2 %
Slices per slab	144
FoV read	180 mm
FoV phase	100.0 %
Slice thickness	1.00 mm
TR	5.00 ms
TE	1.83 ms
Averages	2
Concatenations	1
Filter	Prescan Normalize, Elliptical filter
Coil elements	RBR

Contrast

Flip angle	10.0 deg
Fat suppr.	SPAIR
Lines Per Shot	128
Water suppr.	None
Dixon	No Dixon
Save original images	On
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution

Base resolution	320
Phase resolution	85 %
Slice resolution	61 %
Phase partial Fourier	6/8
Slice partial Fourier	6/8
Interpolation	Off
PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	24
Matrix Coil Mode	Auto (Triple)
Reference scan mode	Integrated
Image Filter	Off

Physio

1st Signal/Mode	None
Resp. control	Off

Inline

3D centric reordering	Off
Subtract	Off
Liver registration	Off
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On
Wash - In	Off
Wash - Out	Off

SIEMENS MAGNETOM TrioTim syngo MR B19

TTP	Off
PEI	Off
MIP - time	Off

Sequence

Introduction	Off
Dimension	3D
Elliptical scanning	Off
Reduce Motion Sens.	On
Asymmetric echo	Weak
Contrasts	1
Bandwidth	420 Hz/Px
Optimization	None
Allowed delay	0 s
RF pulse type	Normal
Gradient mode	Fast
Excitation	Slab-sel.
RF spoiling	On

SIEMENS MAGNETOM TrioTim syngo MR B19

\\USER\BREAST (all on REF)\ROUTINE\dynamic-T2s w/oFS 8/12/15\C 3D T1 sag LEFT

TA: 3:42 PAT: 2 Voxel size: 0.7×0.6×1.0 mm Rel. SNR: 1.00 SIEMENS: fl3d_vibe

Properties

Prio Recon	Off
Before measurement	
After measurement	
Load to viewer	On
Inline movie	Off
Auto store images	On
Load to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Start measurement without further preparation	On
Wait for user to start	On
Start measurements	single

Distortion Corr.	Off
Unfiltered images	Off
Prescan Normalize	On
Normalize	Off
B1 filter	Off
Raw filter	Off
Elliptical filter	On
Mode	Inplane
POCS	Off

Geometry

Multi-slice mode	Sequential
Series	Ascending
Special sat.	None

System

Body	Off
LBR	On
RBR	Off
Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Save uncombined	Off
Coil Combine Mode	Sum of Squares
Auto Coil Select	Default
Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	On
Assume Silicone	Off
? Ref. amplitude 1H	0.000 V
Adjustment Tolerance	Auto
Adjust volume	
Position	L85.9 A22.9 H17.4
Orientation	Sagittal
Rotation	90.00 deg
A >> P	180 mm
F >> H	180 mm
R >> L	144 mm

Routine

Slab group 1	
Slabs	1
Dist. factor	20 %
Position	L85.9 A22.9 H17.4
Orientation	Sagittal
Phase enc. dir.	H >> F
Rotation	90.00 deg
Phase oversampling	100 %
Slice oversampling	22.2 %
Slices per slab	144
FoV read	180 mm
FoV phase	100.0 %
Slice thickness	1.00 mm
TR	5.00 ms
TE	1.83 ms
Averages	2
Concatenations	1
Filter	Prescan Normalize, Elliptical filter
Coil elements	LBR

Contrast

Flip angle	10.0 deg
Fat suppr.	SPAIR
Lines Per Shot	128
Water suppr.	None
Dixon	No Dixon
Save original images	On
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution

Base resolution	320
Phase resolution	85 %
Slice resolution	61 %
Phase partial Fourier	6/8
Slice partial Fourier	6/8
Interpolation	Off
PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	24
Matrix Coil Mode	Auto (Triple)
Reference scan mode	Integrated
Image Filter	Off

Physio

1st Signal/Mode	None
Resp. control	Off

Inline

3D centric reordering	Off
Subtract	Off
Liver registration	Off
Std-Dev-Sag	Off
Std-Dev-Cor	Off
Std-Dev-Tra	Off
Std-Dev-Time	Off
MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On
Wash - In	Off
Wash - Out	Off

SIEMENS MAGNETOM TrioTim syngo MR B19

TTP	Off
PEI	Off
MIP - time	Off

Sequence

Introduction	Off
Dimension	3D
Elliptical scanning	Off
Reduce Motion Sens.	On
Asymmetric echo	Weak
Contrasts	1
Bandwidth	420 Hz/Px
Optimization	None
Allowed delay	0 s
RF pulse type	Normal
Gradient mode	Fast
Excitation	Slab-sel.
RF spoiling	On

Table of contents

\\USER

BREAST (all on REF)

ROUTINE

dynamic-T2s w/oFS 8/12/15

turn OFF auto coil select

ALL series must be at REF

localizer

STIR COR if known CANCER

t1_fl3d_tra_nonfatsat

t2_tse_tra_spair

SIEMENS t1_3d_tra_90s-dynFS - TEST

SIEMENS t1_3d_tra_90s-dynFS - TEST #2

injection

ALL series must be at REF

use MULTIHANCE per DR>FIDLER

SIEMENS t1_3d_tra_90s-dynVIEWS_1+4 ACR

REFORMAT RT/LT BREAST SAG POST

turn ON auto coil select

OPTIONAL fat sat

T2fs SAG RIGHT

T2fs SAG LEFT

T2 SAG RIGHT non fs

T2 SAG LEFT non fs

+C 3D T1 sag RIGHT

+C 3D T1 sag LEFT